Computational Modeling of Bound States in the Yukawa Potential

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Abstract

The Yukawa potential is a short-range interaction proposed by Hideki Yukawa in 1935 to describe the strong nuclear force through the proposal of a massive particle, later identified as the meson. His proposal explains how the meson mediates the attractive force between nucleons and overcomes the repulsive electromagnetic force at short distances, contributing to the stability of atomic nuclei. This model acts as a screened Coulomb potential with a similar $1/r$ dependence and acts over distances on the order of 10^{-15} m (1 fm), but with an exponential screening factor of $e^{-\lambda r}$. We aim to build a computational solver exploring bound and pseudo-bound states by solving the non-relativistic radial Schrödinger equation. The main focus is to solve for low-energy bound and pseudo-bound state eigenenergies and eigenfunctions using numerical methods and to plot the corresponding wavefunctions as functions of radial distance r. These bound and pseudo-bound states form and are dependent on varying system parameters, such as potential well depth and width, as well as screening strength. For example, as the potential well depth increases, more bound states appear. We aim to examine how these states depend on the strength and range of the parameters, and identify the conditions under which bound and pseudo-bound states emerge and disappear.